

Total No. of Questions : 8]

SEAT No. :

PC2428

[Total No. of Pages : 2

[6354]-549

B.E. (E & TC)

REMOTE SENSING

(2019 Pattern) (Semester - VIII) (Elective - VI) (404192C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.

- Q1) a)** Write a list of three categories of satellite data products. With help of diagrams and suitable example explain any one satellite data product. **[8]**
- b) Explain the preprocessing techniques applied on satellite imagery in remote sensing? **[8]**

OR

- Q2) a)** Explain the elements of visual interpretation used in remote sensing with suitable illustrations. **[8]**
- b) Elaborate the major differences between NDVI, NDWI and NDSI with its equations. **[8]**
- Q3) a)** Explain microwave remote sensing, its types and its advantages over optical remote sensing. **[8]**
- b) Explain principles of radar, its equation and derive range resolutions of SLAR. **[10]**

OR

- Q4) a)** Explain the principle of altimeter with a suitable diagram and its applications. **[8]**
- b) Explain various instruments / sensors used in microwave remote sensing with suitable examples. **[10]**

P.T.O.

- Q5) a)** Explain the working principle behind the GNSS technology, its features and applications. **[10]**
- b)** Explain the vulnerabilities of GNSS and its signal structure. **[8]**

OR

- Q6) a)** Explain ground based augmentation system and space based augmentation system. **[10]**
- b)** Explain the concept of Differential GPS and Location Based Services and its use cases. **[8]**

- Q7) a)** Brief the application of remote sensing in development of Land Information System with use cases and methodology. **[10]**
- b)** Brief the application of remote sensing in coastal and near sea shore management with use cases and methodology. **[8]**

OR

- Q8) a)** Brief the application of remote sensing in managing Smart cities with use cases and methodology. **[10]**
- b)** Brief the application of remote sensing in natural disasters monitoring and mitigating with use cases and methodology. **[8]**

